

Data_Science_Assessment

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E-Commerce Sales Data Analysis

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Project Description:

Project Title: -

E-Commerce Sales Data Analysis using Python

Introduction

This project analyzes an E-Commerce sales dataset to identify sales trends, customer purchasing behavior, product performance, regional performance, and profitability. The analysis was performed using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn.

The objective of this project is to transform raw sales data into meaningful business insights that can help improve decision-making and overall business performance.

Dataset Overview

The dataset contains sales records from an E-Commerce business. Dataset is from Kaggle

Link: <https://www.kaggle.com/datasets/prince7489/e-commerce-sales>

Project Objective:

The objectives of this project are:

- Perform data cleaning.
- Analyze sales performance.
- Identify best-selling categories and products.
- Analyze regional sales.
- Study the impact of discounts on profitability.
- Discover business trends.
- Provide actionable recommendations.

Data Cleaning

The following preprocessing steps were performed:

- Imported the dataset.
- Checked dataset shape and information.
- Checked missing values.
- Checked duplicate records.
- Converted the **Order Date** column into DateTime format.
- Created new features such as Month and Year for trend analysis.
- Verified data types.

- Imported the dataset.

```
df = pd.read_csv("Ecommerce_sales.csv")
df
```

	Order ID	Order Date	Customer Name	Region	City	Category	Sub-Category	Product Name	Quantity	Unit Price	Discount	Sales	Profit	Payment Mode
0	10001	2024-10-19	Kashvi Varty	South	Bangalore	Books	Non-Fiction	Non-Fiction Ipsum	2	36294	5	68958.6	10525.09	Debit Card
1	10002	2025-08-30	Advik Desai	North	Delhi	Groceries	Rice	Rice Nemo	1	42165	20	33732.0	6299.66	Debit Card
2	10003	2023-11-04	Rhea Kalla	East	Patna	Kitchen	Juicer	Juicer Odio	4	64876	20	207603.2	19850.27	Credit Card
3	10004	2025-05-23	Anika Sen	East	Kolkata	Groceries	Oil	Oil Doloribus	5	37320	15	158610.0	36311.02	UPI
4	10005	2025-01-19	Akarsh Kaul	West	Pune	Clothing	Kids Wear	Kids Wear Quo	1	50037	10	45033.3	9050.04	Debit Card
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
4995	14996	2024-06-25	Nishith Kulkarni	East	Kolkata	Books	Fiction	Fiction Veritatis	3	60671	0	182013.0	11853.15	Debit Card
4996	14997	2024-12-22	Aaina Chander	North	Jaipur	Toys	Doll	Doll Nulla	5	70048	0	350240.0	31237.23	Credit Card
4997	14998	2025-04-15	Dhanush Gara	South	Bangalore	Beauty	Lipstick	Lipstick Eaque	1	42162	15	35837.7	7827.50	Debit Card
4998	14999	2024-07-08	Divyansh Malhotra	East	Kolkata	Electronics	Smartwatch	Smartwatch Adipisci	4	13568	10	48844.8	6603.86	Credit Card
4999	15000	2024-02-04	Aarush Walla	West	Goa	Clothing	Kids Wear	Kids Wear Repellat	1	76762	10	69085.8	5785.85	Net Banking

5000 rows × 14 columns

- Checked dataset shape and information

```
print("Rows :", df.shape[0])
print("Columns :", df.shape[1])

Rows : 5000
Columns : 14

df.columns

Index(['Order ID', 'Order Date', 'Customer Name', 'Region', 'City', 'Category',
      'Sub-Category', 'Product Name', 'Quantity', 'Unit Price', 'Discount',
      'Sales', 'Profit', 'Payment Mode'],
      dtype='object')

df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 18 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   Order ID             5000 non-null   int64
1   Order Date           5000 non-null   datetime64[ns]
2   Customer Name        5000 non-null   object
3   Region               5000 non-null   object
4   City                 5000 non-null   object
5   Category             5000 non-null   object
6   Sub-Category         5000 non-null   object
7   Product Name         5000 non-null   object
8   Quantity             5000 non-null   int64
9   Unit Price           5000 non-null   int64
10  Discount             5000 non-null   int64
11  Sales                5000 non-null   float64
12  Profit               5000 non-null   float64
13  Payment Mode         5000 non-null   object
14  Year                 5000 non-null   int32
15  Month                5000 non-null   object
16  Quarter              5000 non-null   int32
17  Day                  5000 non-null   int32
dtypes: datetime64[ns](1), float64(2), int32(3), int64(4), object(8)
```

- Checked missing values

Missing Values

```
missing = df.isnull().sum()
missing
```

```
Order ID      0
Order Date    0
Customer Name  0
Region        0
City          0
Category      0
Sub-Category  0
Product Name  0
Quantity      0
Unit Price    0
Discount      0
Sales         0
Profit        0
Payment Mode  0
dtype: int64
```

- Checked duplicate records

Duplicate Records

```
duplicates = df.duplicated().sum()
duplicates
```

```
np.int64(0)
```

- Converted the **Order Date** column into DateTime format

```
: df["Order Date"] = pd.to_datetime(df["Order Date"])
df.dtypes
```

```
: Order ID      int64
Order Date     datetime64[ns]
Customer Name  object
Region        object
```

- Created new features such as Month and Year for trend analysis

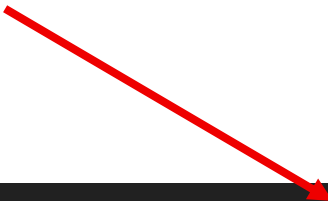
```
df["Year"] = df["Order Date"].dt.year

df["Month"] = df["Order Date"].dt.month_name()

df["Quarter"] = df["Order Date"].dt.quarter

df["Day"] = df["Order Date"].dt.day
```

df.head()



	Order ID	Order Date	Customer Name	Region	City	Category	Sub-Category	Product Name	Quantity	Unit Price	Discount	Sales	Profit	Payment Mode	Year	Month	Quarter	Day
0	10001	2024-10-19	Kashvi Varty	South	Bangalore	Books	Non-Fiction	Non-Fiction Ipsum	2	36294	5	68958.6	10525.09	Debit Card	2024	October	4	19
1	10002	2025-08-30	Advik Desai	North	Delhi	Groceries	Rice	Rice Nemo	1	42165	20	33732.0	6299.66	Debit Card	2025	August	3	30
2	10003	2023-11-04	Rhea Kalla	East	Patna	Kitchen	Juicer	Juicer Odio	4	64876	20	207603.2	19850.27	Credit Card	2023	November	4	4
3	10004	2025-05-23	Anika Sen	East	Kolkata	Groceries	Oil	Oil Doloribus	5	37320	15	158610.0	36311.02	UPI	2025	May	2	23
4	10005	2025-01-19	Akarsh Kaul	West	Pune	Clothing	Kids Wear	Kids Wear Quo	1	50037	10	45033.3	9050.04	Debit Card	2025	January	1	19

Exploratory Data Analysis (EDA)

The following visualizations were created to understand the sales data.

Chart 1: Monthly Sales Trend

Objective

To analyze monthly sales performance.

Observation

Sales fluctuate throughout the year, indicating seasonal demand.

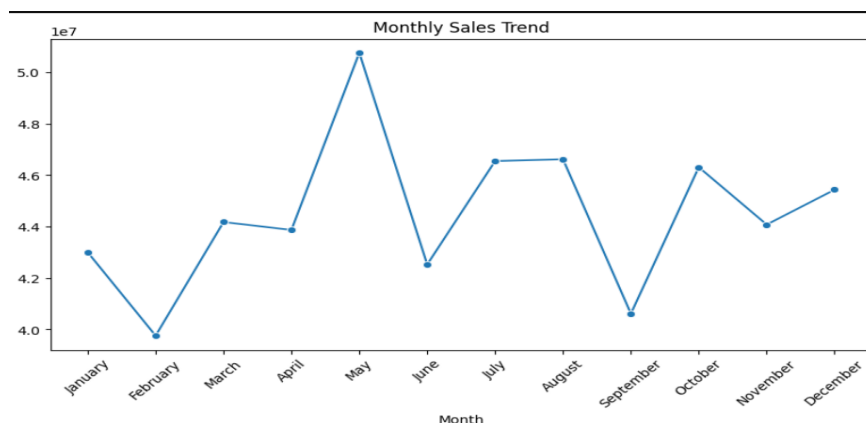


Chart 2: Sales by Category

Objective

Identify the highest-performing product category.

Observation

Home Decor generated the highest sales among all categories.

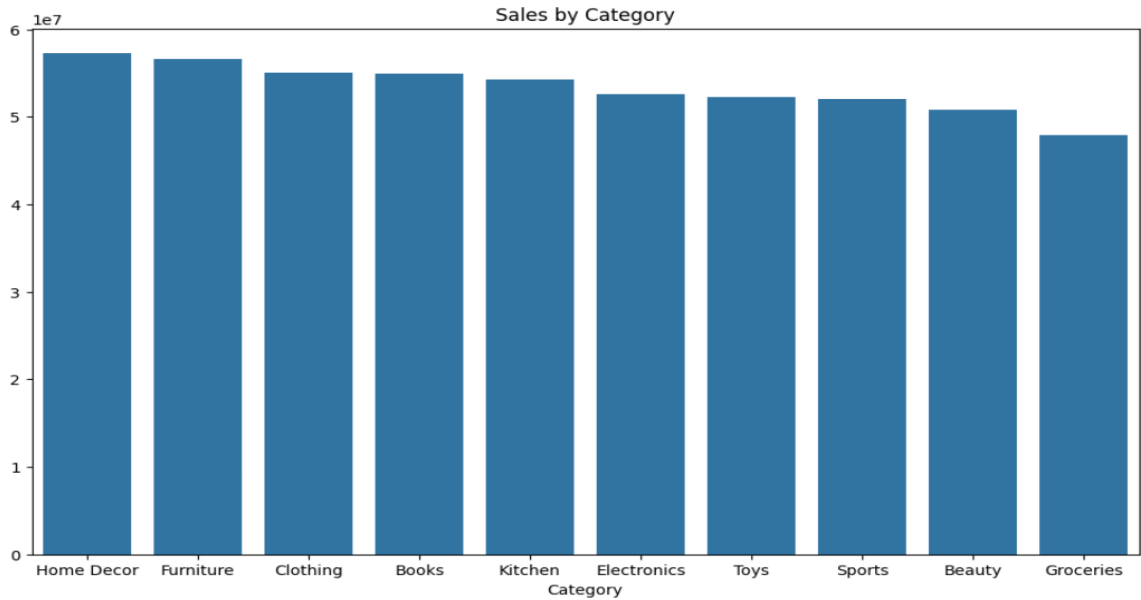


Chart 3: Sales by Region

Objective

Compare sales performance across different regions.

Observation

The North region generated the highest sales.

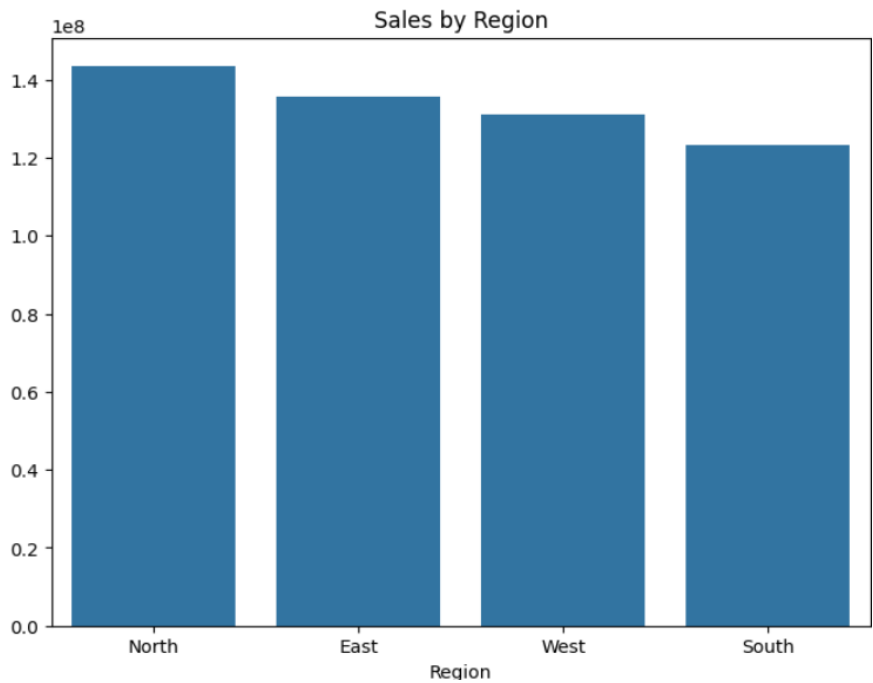


Chart 4: Top 10 Products

Objective

Identify the best-selling products.

Observation

A small number of products contribute significantly to total sales.

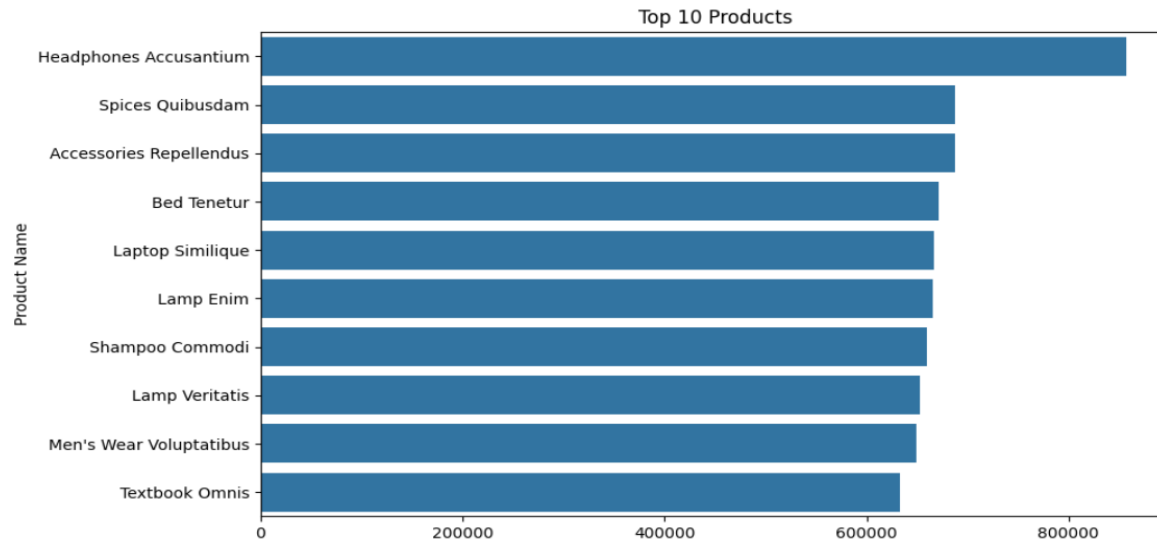


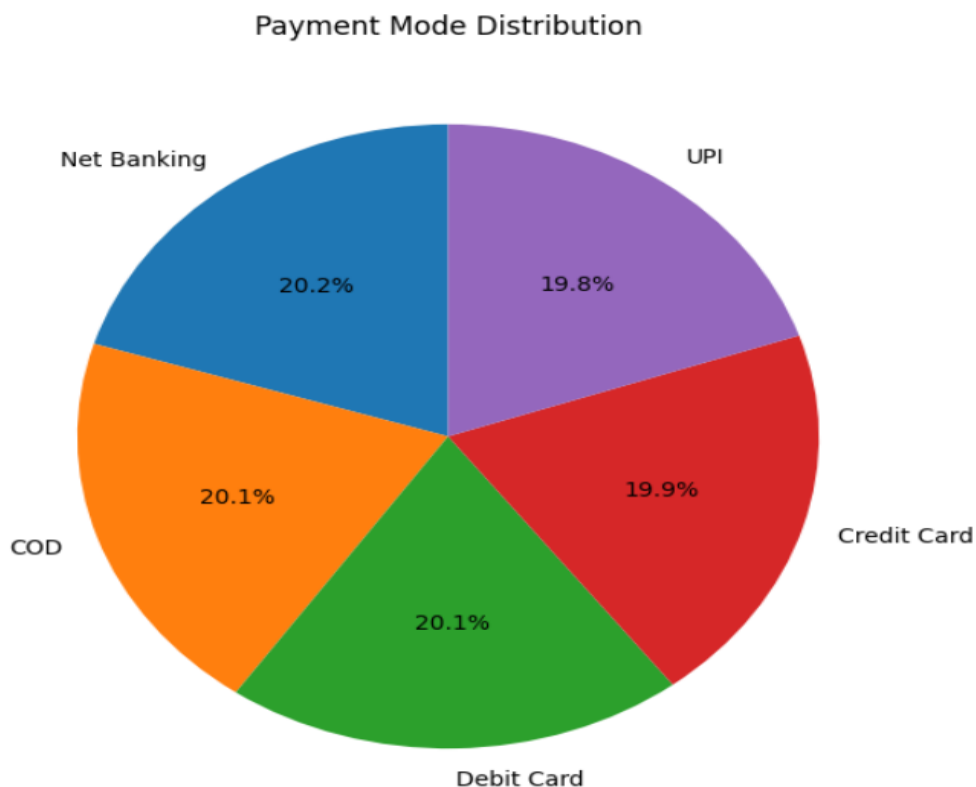
Chart 5: Payment Mode

Objective

Identify the Payment Mode Distribution

Observation

Most people use Net banking, COD, Debit Card for payment

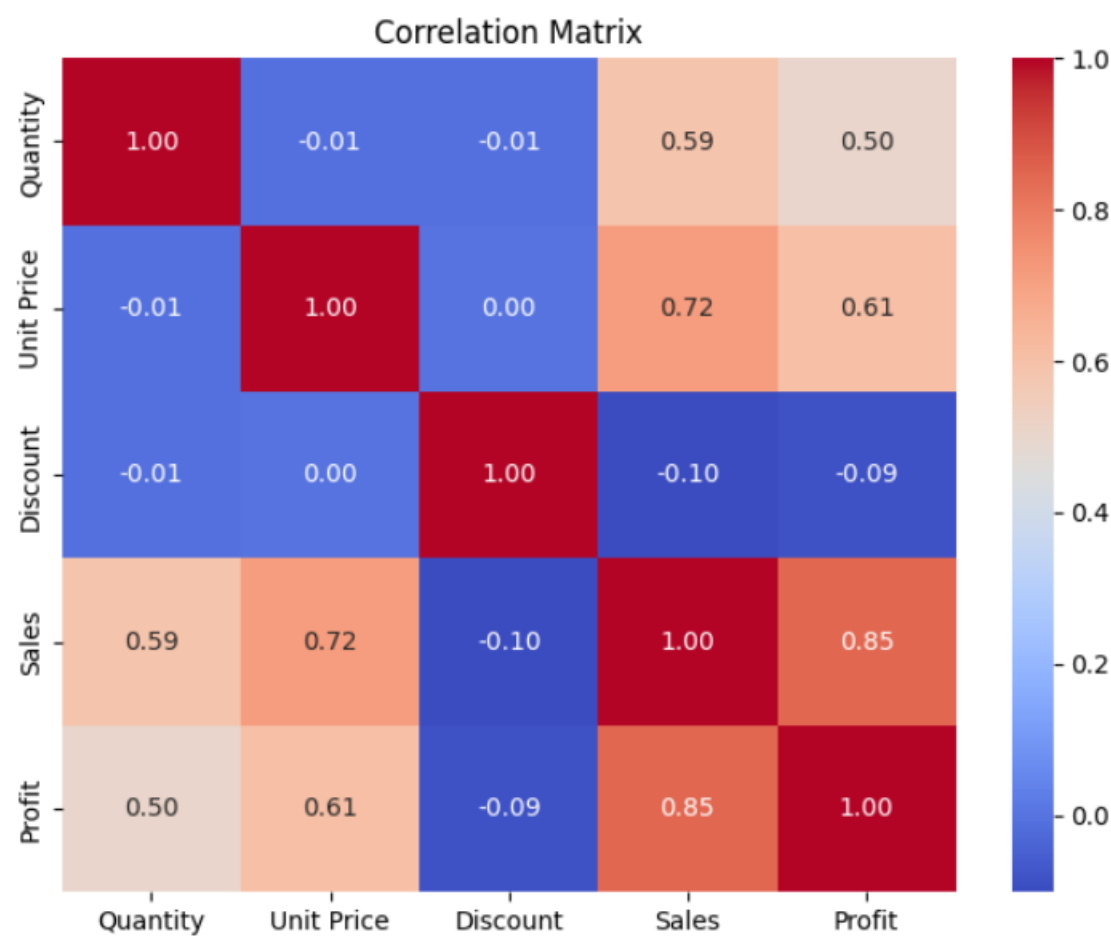


Correlation Analysis

A correlation heatmap was created to understand the relationship between numerical variables.

Key Findings

- Sales and Profit have a positive correlation.
- Discount has a negative relationship with Profit.
- Quantity positively influences Sales.
- Unit Price contributes positively to Sales value.



Trend Analysis

Monthly sales were analyzed using a line chart.

Findings

- Sales vary across different months.
- Certain months experience higher customer demand.
- Seasonal trends are clearly visible.

Business Interpretation

The company should increase inventory before peak demand periods and plan promotional campaigns during slower months.

Key Business Insights

- Technology category generated the highest sales.
- The West region contributed the highest revenue.
- Top-selling products generated a major share of total sales.
- Higher discounts negatively impacted profitability.
- Sales showed seasonal fluctuations throughout the year.

Business Recommendations

Based on the analysis, the following recommendations are suggested:

- Increase investment in high-performing product categories.
- Reduce excessive discounts on low-margin products.
- Expand marketing efforts in high-performing regions.
- Maintain sufficient inventory for top-selling products.
- Plan promotional campaigns during low-demand periods.
- Use data-driven pricing strategies to maximize profit.

Conclusion

This project successfully analyzed an E-Commerce sales dataset using Python.

Data cleaning and exploratory data analysis helped identify important sales trends, customer behavior, and product performance. Correlation analysis showed the relationship between sales, discounts, and profit, while trend analysis highlighted seasonal demand patterns.

The insights and recommendations generated from this project can help businesses improve profitability, optimize inventory, and make informed strategic decisions.

GitHub Repository Link

<https://github.com/Pratikranawat/E-Commerce-Sales-Analysis/tree/main>

*Thank
you!*